Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

In an adiabatic process, the pressure of a monoatomic ideal gas increases by 0.5%. The volume will decrease by

(A) 0.1%

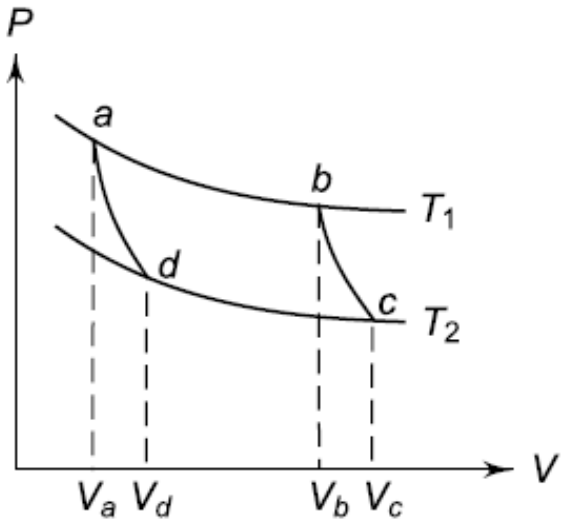
(B) 0.2%

(C) 0.3%

(D) 0.4%

Q.2

Two different adiabatic paths for the same gas intersect two isothermals at T_1 and T_2 as shown in the $P-V$ diagram (Fig.). Then



(A) $\frac{V_a}{V_c}=\frac{V_b}{V_d}$

(B) $\frac{V_a}{V_b}=\frac{T_2}{T_1}$

(C) $\frac{V_a}{V_b}=\frac{V_d}{V_c}$

(D) $\frac{V_a}{V_d}=\frac{T_1}{T_2}$

Q.3

An ideal gas is expanded isothermally from a volume V_1 to volume V_2 and then compressed adiabatically to original volume V_1. The initial pressure is P_1 and the final pressure is P_3. If the net work done is W then

(A) $P_3>P_1, W>0$

(B) $P_3<P_1, W<0$

@IITJEE_Advanced

(C) $P_3 > P_1, W < 0$

(D) $P_3 = P_1, W = 0$

Q.4

During an adiabatic process, the pressure of a gas is proportional to the cube of its absolute temperature. The value of C_p/C_v for that gas is:

(A) $\frac{3}{5}$

(B) $\frac{4}{3}$

(C) $\frac{5}{3}$

(D) $\frac{3}{2}$

Q.5

A gas does $4.5\ \text{J}$ of external work during adiabatic expansion. Its temperature falls by $2\ \text{K}$. Its internal energy will

(A) increase by $4.5\ \text{J}$

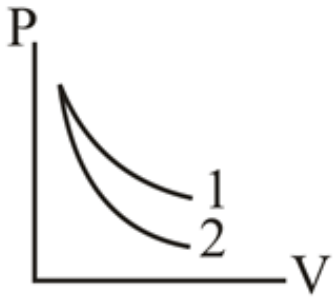
(B) decrease by $4.5\ \text{J}$

(C) decrease by $2.25\ \text{J}$

(D) increase by $9.0\ \text{J}$

Q.6

The *P* - *V* plots for two gases during adiabatic processes are shown in the figure. Plots 1 and 2 should correspond, respectively, to,



(A) He and O_2.

(B) O_2 and He.

(C) He and Ar.

(D) O_2 and N_2.

Q.7

Starting with the same initial conditions, an ideal gas expands from volume V_1 to V_2 in three different ways.

@IITJEE_Advanced

The work done by the gas is W_1 if the process is purely isothermal, W_2 if purely isobaric and W_3 is purely adiabatic. Then

(A) $W_2 > W_1 > W_3$

(B) $W_2 > W_3 > W_1$

(C) $W_1 > W_2 > W_3$

(D) $W_1 > W_3 > W_2$

Q.8

Two identical containers A and B fitted with frictionless pistons contains the same ideal gas at the same temperature and the same volume V. The mass of the gas in A is m_A and that in B is m_B. The gas in each cylinder is now allowed to expand isothermally to the same final volume $2V$. The changes in pressure in A and B are found to be ΔP and $1.5\Delta P$, respectively. Then

(A) $4m_A = 9m_B$

(B) $2m_A = 3m_B$

(C) $3m_A = 2m_B$

(D) $9m_A = 4m_B$

Q.9

Two cylinders A and B fitted with pistons contain equal amounts of an ideal diatomic gas at $300\ \mathrm{K}$. The piston of A is free to move, while that of B is held fixed. The same amount of heat is given to the gas in each cylinder. If the rise in temperature of the gas in A is $30\ \mathrm{K}$, then the rise in temperature of the gas in B is

(A) $30\ \mathrm{K}$

(B) $18\ \mathrm{K}$

(C) $50\ \mathrm{K}$

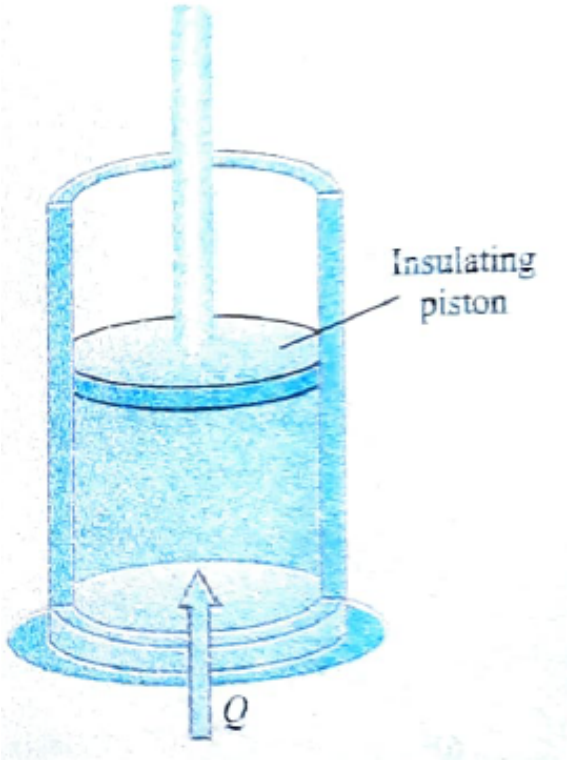
(D) $42\ \mathrm{K}$

Q.10

An adiabatic vertical cylinder fitted with and adiabatic frictionless piston contains n moles of an ideal diatomic gas. The initial volume and temperature of the gas are V_0 and T_0 respectively, with the piston fixed tightly. The atmospheric pressure is P_0. If the piston be released, find the equitibrium temperature and volume of the gas, if the heat capacity of the cylinder and piston is negligibly small. Weight of the piston and cross-sectional area of

@IITJEE_Advanced

the cylinder are W and A respectively.



(A) $\frac{2V_0}{q} + \frac{5nRT_0}{9\left(P_0+\frac{W}{A}\right)}$

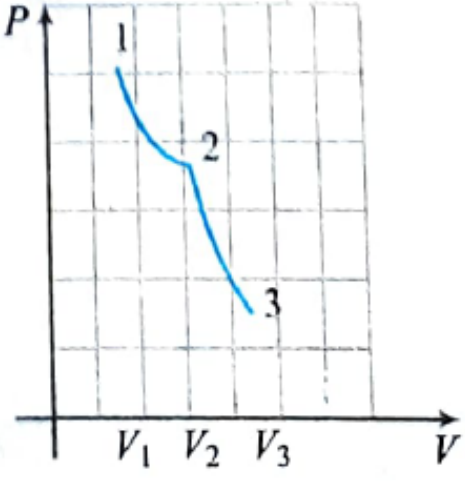
(B) $\frac{2V_0}{5} + \frac{7nRT_0}{5\left(P_0+\frac{W}{A}\right)}$

(C) $\frac{2V_0}{7} + \frac{5nRT_0}{7\left(P_0+\frac{W}{A}\right)}$

(D) $\frac{3V_0}{2} + \frac{5nRT_0}{2\left(P_0+\frac{W}{A}\right)}$

Q.11

One mole of a perfect gas, initially at a pressure and temperature of $10^5\ \mathrm{N/m^2}$ and $300\ \mathrm{K}$, respectively, expands isothermally until its volume is doubled and then adiabatically until its volume is again doubled. Find the final pressure and temperature of the gas. Find the total work done during the isothermal and adiabatic processes. Given $\gamma = 1.4$. Also draw the $P - V$ diagram for the process.



(A) 3233.56 J

(B) 3900.24 J

(C) 3444.78 J

@IITJEE_Advanced

(D) 5673.86 J

Q.12

When two moles of oxygen is heated from 0°C to 10°C at constant volume, its internal energy changes by $420\ \text{J}$. What is the molar specific heat of oxygen at constant volume?

(A) $5.75\ \text{J K}^{-1}\ \text{mol}^{-1}$

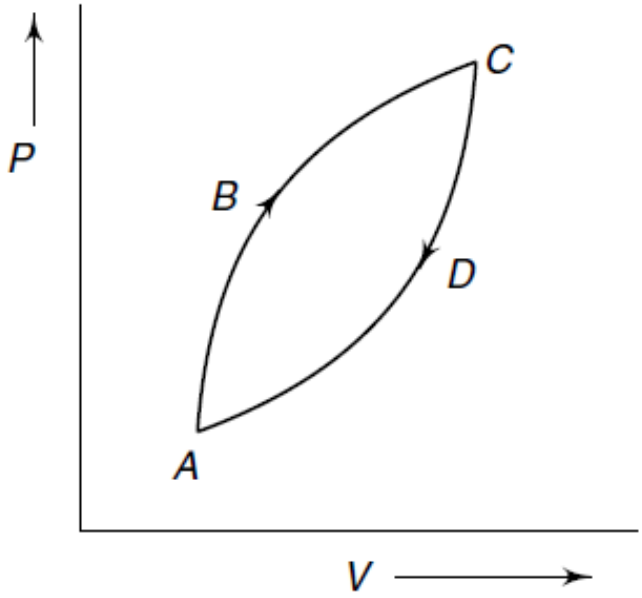
(B) $10.5\ \text{J K}^{-1}\ \text{mol}^{-1}$

(C) $21\ \text{J K}^{-1}\ \text{mol}^{-1}$

(D) $42\ \text{J K}^{-1}\ \text{mol}^{-1}$

Q.13

Figure shows the $P - V$ diagram of a cyclic process. If dQ is the heat energy supplied to the system, dU is the change in internal energy of the system and dW is the work done by the system, then which of the following relations is correct?



(A) $dQ = dU - dW$

(B) $dU = 0$

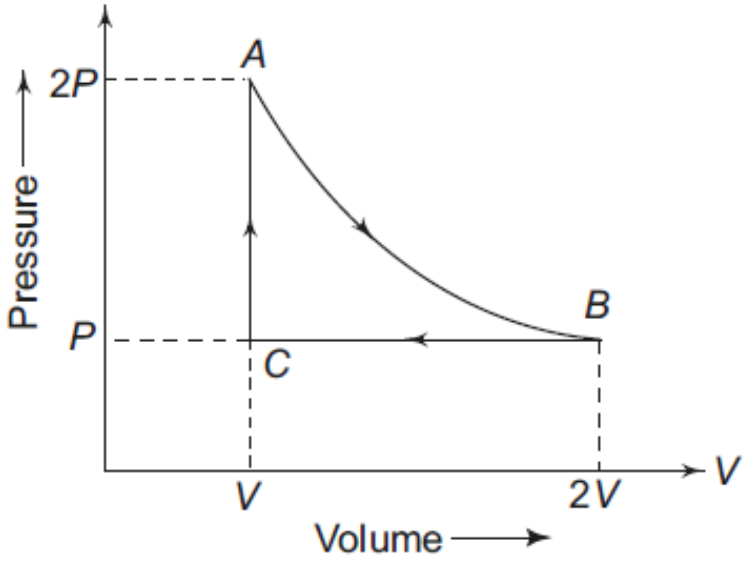
(C) $dU = dW$

(D) $dQ = -dW$

Q.14

@IITJEE_Advanced

In Fig., if the temperature at A is T, what is the temperature at C?



(A) $4T$

(B) $2T$

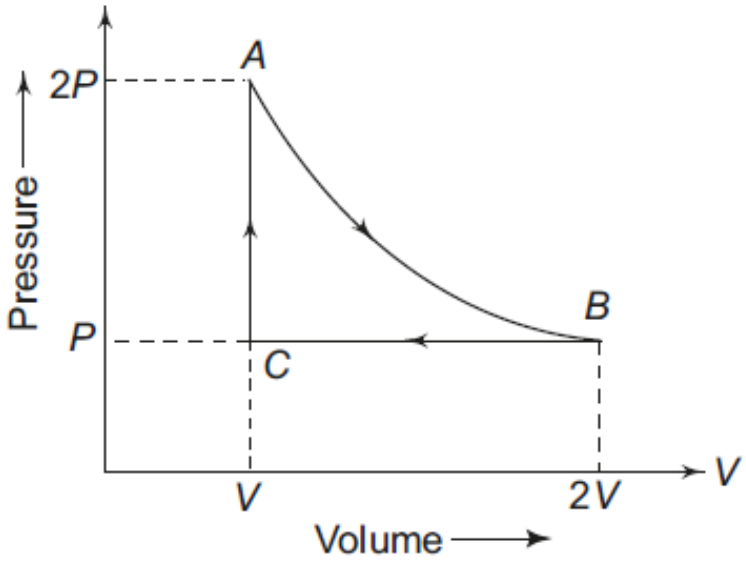
(C) T

(D) $\frac{T}{2}$

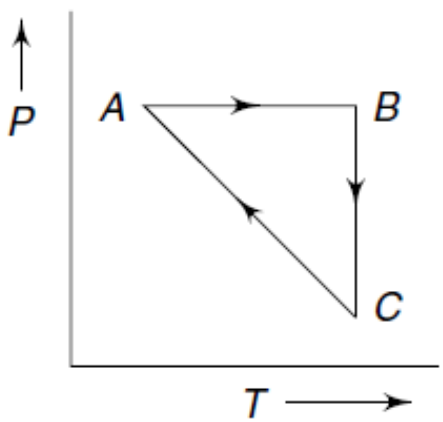
Q.15

Figure shows the $P-V$ diagram for a fixed mass of an ideal gas undergoing cyclic process $ABCA$. AB represents isothermal process.

Which of the graphs shown in Fig. represents the $P-T$ diagram of the cyclic process?

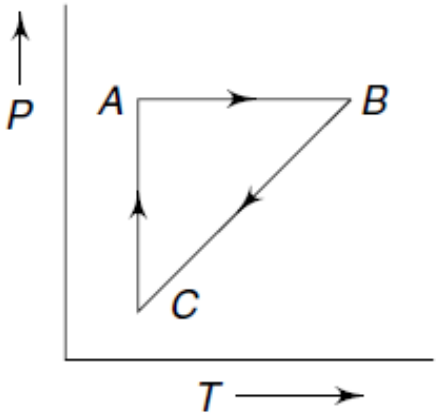


(A)

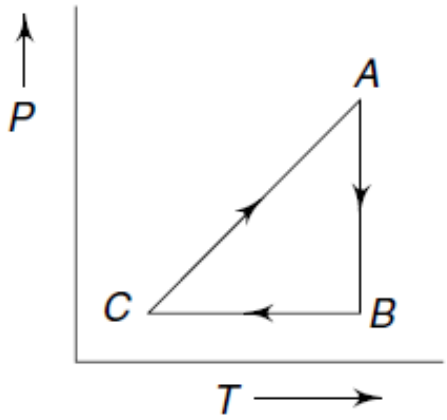


@IITJEE_Advanced

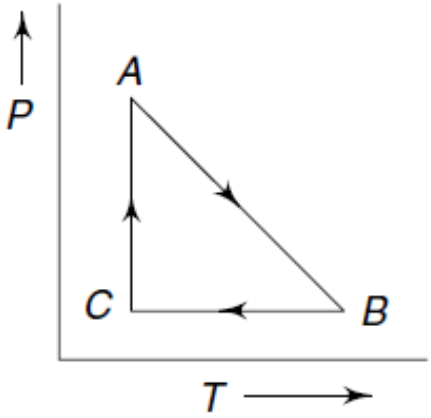
(B)



(C)



(D)



Q.16

In a cylinder, 2.0 moles of an ideal monatomic gas initially at 1.0×10^6 Pa and 300 K expands until its volume doubles. Compute the work done if the expansion is (a) isothermal, (b) adiabatic and (c) isobaric.

(A) 3452 J, 2764 J, 4980 J

(B) 3402 J, 2764 J, 4900 J

(C) 3472 J, 2664 J, 4980 J

(D) 5673 J, 3456 J, 3457 J

Q.17

When an ideal gas at pressure P, temperature T and volume V is isothermally compressed to V/n, its pressure becomes P_i. If the gas is compressed adiabatically to V/n, its pressure becomes P_a. The ratio P_i/P_a is $(\gamma = C_p/C_v)$

@IITJEE_Advanced

(A) 1

(B) n

(C) n^{γ}

(D) $n^{(1-\gamma)}$

Q.18

The ends Q and R of two thin wires PQ and RS are soldered (joined) together. Initially each of the wires has a length of $1\ \mathrm{m}$ at $10^{\circ}\mathrm{C}$. Now the end P is maintained at $10^{\circ}\mathrm{C}$, while the end S is heated and maintained at $400^{\circ}\mathrm{C}$. The system is thermally insulated from its surroundings. If the thermal conductivity of the wire PQ is twice that of the wire RS, and the coefficient of linear thermal expansion for PQ is $1.2 \times 10^{-5}\ \mathrm{K}^{-1}$, the change in the length of the wire PQ is

(A) $0.78\ \mathrm{mm}$

(B) $0.90\ \mathrm{mm}$

(C) $1.56\ \mathrm{mm}$

(D) $2.34\ \mathrm{mm}$

Q.19

An ideal Carnot's engine works between $227\ \ ^{\circ}\mathrm{C}$ and $57\ \ ^{\circ}\mathrm{C}$. The efficiency of the engine will be

(A) 22%

(B) 34%

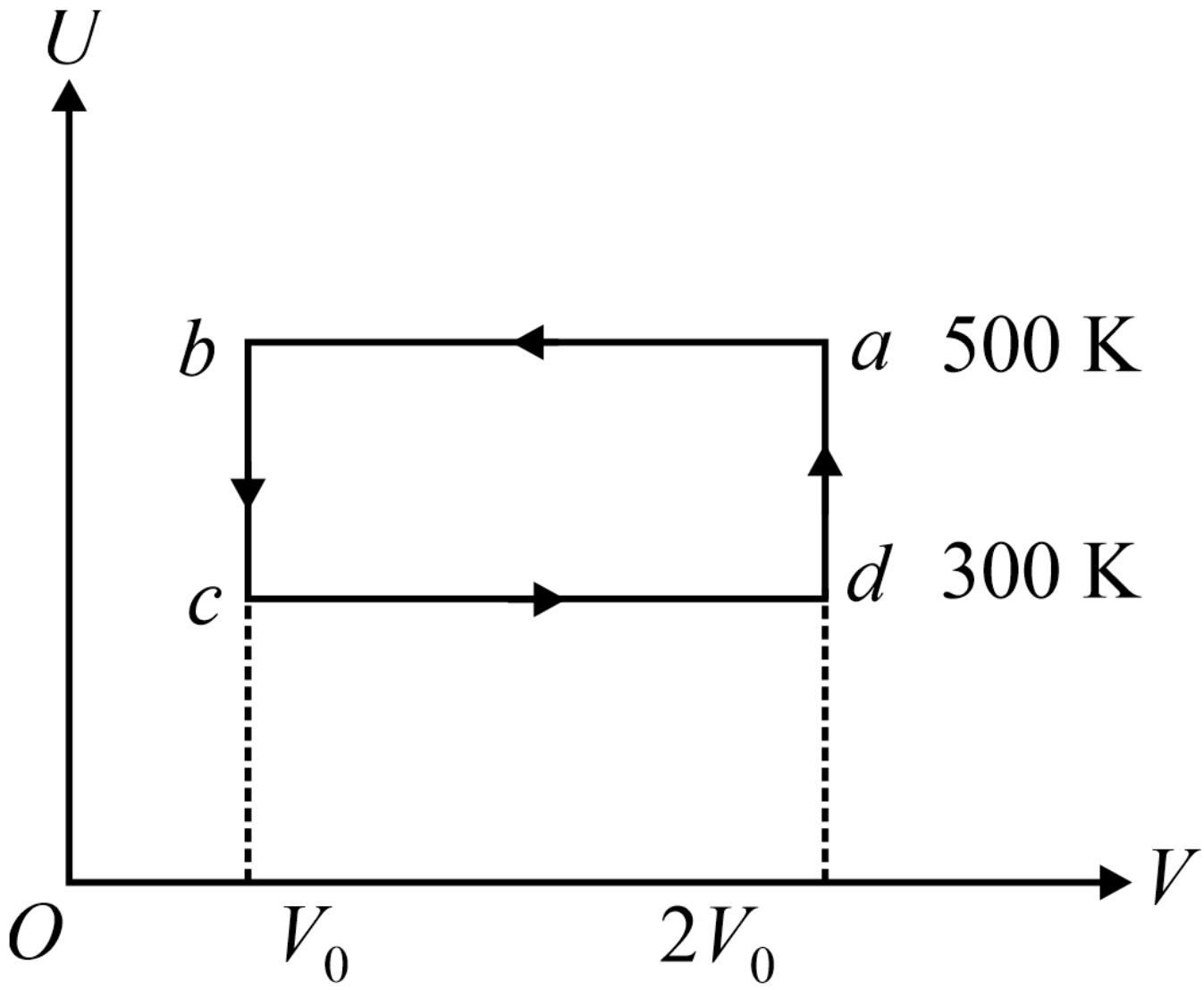
(C) 55%

(D) 13.5%

Q.20

The diagram given below shows the variation in the internal energy U with the volume V of 2.0 moles of an ideal gas in cyclic process abcda. The temperatures of the gas at b and c are $500\ \ \mathrm{K}$ and $300\ \ \mathrm{K}$, respectively. The heat absorbed by the gas during the process is

@IITJEE_Advanced



(A) $400R\ln(2)$

(B) $500R\ln(2)$

(C) $700R\ln(2)$

(D) $800R\ln(2)$

Answers & Solutions

Q.1 Answer:

0.3%

Solution:

In a adiabatic process $PV^{\gamma} =$ constant (k). Taking logarithm,

$\log P + \gamma \log V = \log k$

Differentiating,

$\frac{\Delta P}{P} + \gamma \frac{\Delta V}{V} = 0$

$\Rightarrow \frac{\Delta V}{V} = -\frac{1}{\gamma}\frac{\Delta P}{P}$

For a monoatomic gas, $\gamma = \frac{5}{3}$. Therefore

@IITJEE_Advanced

$\frac{\Delta V}{V} = -\frac{3}{5} \times 0.5\% = -0.3\%$

The negative sign shows that the volume decreases. So the correct choice is (c).

Q.2 Answer:

$\frac{V_a}{V_b} = \frac{V_d}{V_c}$

Solution:

The two adiabatic paths ad and bc for the gas intersect the two isothermals ab and cd at temperatures T_1 and T_2. Since points a and d lie on the same adiabatic path, we have

$$T_1 V_a^{(\gamma-1)} = T_2 V_d^{(\gamma-1)}$$

$$\text{or } \left(\frac{V_a}{V_d}\right)^{(\gamma-1)} = \frac{T_2}{T_1} \quad (1)$$

Since points b and c also lie on the same adiabatic path,

$$T_1 (V_b)^{(\gamma-1)} = T_2 V_c^{(\gamma-1)}$$

$$\text{or } \left(\frac{V_b}{V_c}\right)^{(\gamma-1)} = \frac{T_2}{T_1} \quad (2)$$

From (1) and (2), we get

$$\left(\frac{V_a}{V_d}\right)^{(\gamma-1)} = \left(\frac{V_b}{V_c}\right)^{(\gamma-1)}$$

$$\text{or} \quad \frac{V_a}{V_d} = \frac{V_b}{V_c}$$

Thus the correct choice is (c).

Q.3 Answer:

$P_3 > P_1, W < 0$

Solution:

For an isothermal process: $PV =$ constant and for an adiabatic process: $PV^\gamma =$ constant, where γ is the ratio of the two specific heats (C_p/Cv) of the gas. When a gas is compressed from a volume V to a volume $(V - \Delta V)$, the increase in pressure is

$(\Delta P)_{\text{adia}} = \frac{\gamma \Delta V P}{V}$ for an adiabatic compression

and $(\Delta P)_{\text{iso}} = \frac{\Delta V P}{V}$ for an isothermal compression.

Hence P_3 will be greater than P_1. Therefore, the $P - V$ diagrams of isothermal expansion from V_1, P_1 to V_2, P_2 and adiabatic compression for V_2, P_2 to V_1, P_3 are as shown in Fig. 8.38. Let W_1 and W_2 be the work done in isothermal expansion and adiabatic compression respectively.

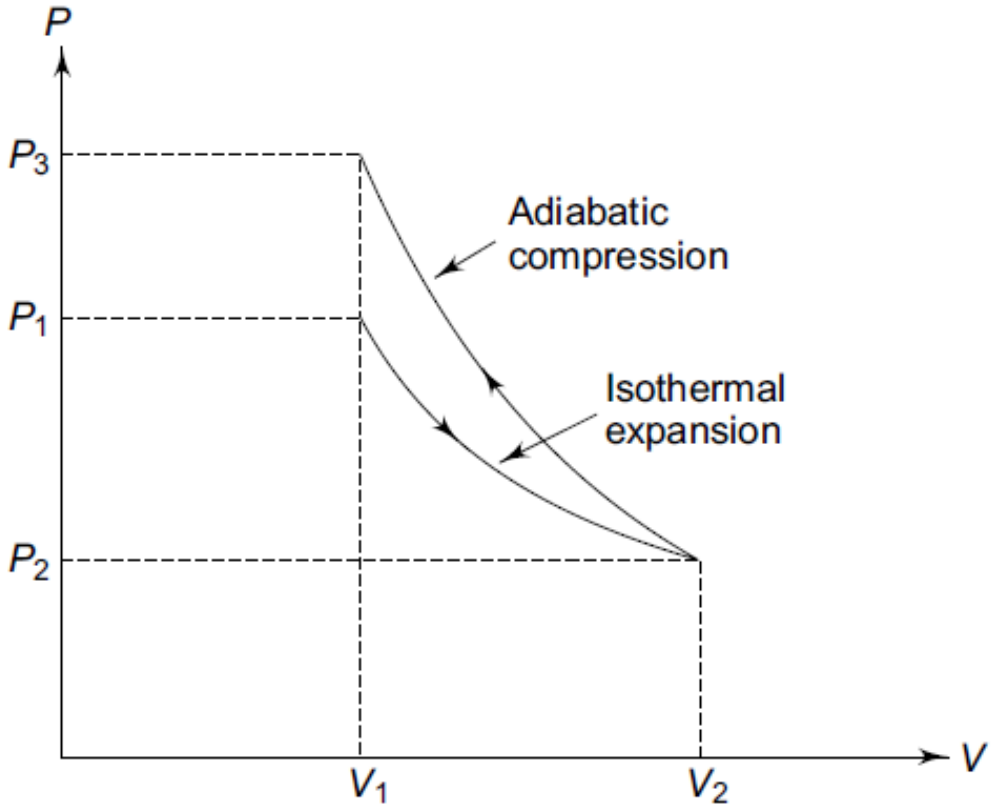
Therefore, net work done is

$W = W_1 + (-W_2) = W_1 - W_2$

Now, the area under the adiabatic curve is more than that under the isothermal curve. Hence $W_2 >$

@IITJEE_Advanced

W_1. **Therefore,** $W < 0$. **Hence the correct choice is (c).**



Q.4 Answer:

$\frac{3}{2}$

Solution:

For an adiabatic process $TP^n = k$ **where** $n = \frac{(1-\gamma)}{\gamma}$, $\gamma = \frac{C_p}{C_v}$ **and** k **is a constant. Hence** $P = \left(\frac{k}{T}\right)^{1/n}$. **Since** $n =$ **constant for a given gas, we have** $P \propto T^{-1/n}$.
Given $P \propto T^3$. **Hence** $-\frac{1}{n} = 3$
$-\frac{\gamma}{1-\gamma} = 3$ **which gives** $\gamma = \frac{3}{2}$ **which is choice (d).**

Q.5 Answer:

decrease by 4.5 J

Solution:

$dQ = dU + dW$. **In an adiabatic change** $dQ = 0$. **Hence,** $dU = -dW = -4.5$ J. **Thus, the correct choice is (b).**

Q.6 Answer:

O_2 and He.

Solution:

For Adiabatic process,
$PV^\gamma =$ constantand $\frac{dP}{dV} = -\gamma P$
$slope_2 > slope_1 \quad \Rightarrow \gamma_2 > \gamma_1$
Also $\gamma_{O_2} > \gamma_{He}$.

@IITJEE_Advanced

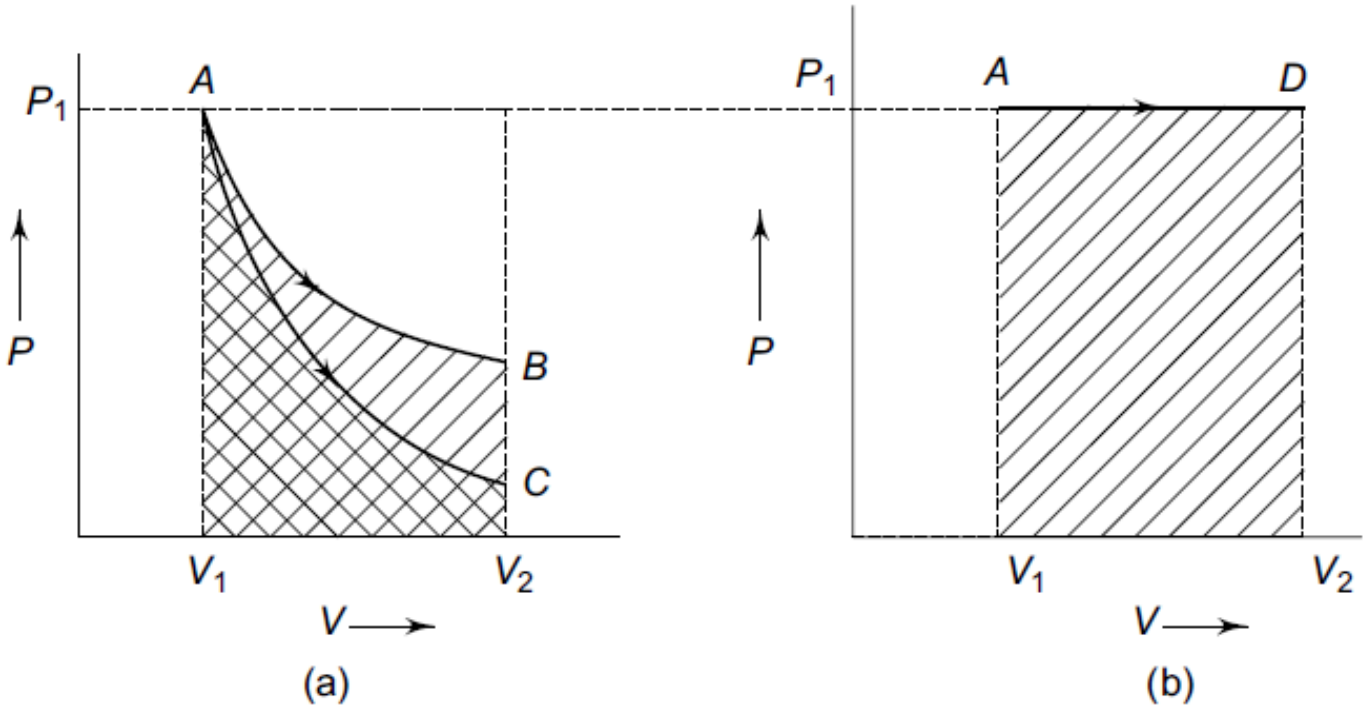
Q.7 Answer:

$W_2 > W_1 > W_3$

Solution:

Since the slope of the $P - V$ graph for adiabatic expansion is γ times that for isothermal expansion, curves AB and AC in Fig. (a) respectively represent isothermal and adiabatic expansions of the gas from initial volume V_1 to final volume V_2.

As the area under curve AB between volumes V_1 and V_2 is greater than the area under curve AC between V_1 and V_2, it follows that $W_1 > W_3$. Figure (b) shows the $P - V$ graph for isobaric (at constant pressure) expansion from initial volume V_1 and pressure P_1 to final volume V_2; the pressure remaining unchanged at P_1. Comparing figures (a) and (b) we find that the area under AD between volumes V_1 and V_2 is greater than the area under curves AB and AC. Hence W_2 is greater than W_1 and W_3. Hence the correct choice is (a).



Q.8 Answer:

$3m_A = 2m_B$

Solution:

The equation of state for an ideal gas of mass m and molecular mass M is

$PV = \frac{m}{M}RT$ (i)

For an isothermal process, $T =$ constant. Differentiating (i) partially at constant T, we get

$$P\Delta V + V\Delta P = 0$$

$$\text{or } \Delta P = -P\frac{\Delta V}{V} \quad \text{(ii)}$$

From (i), $P = \frac{mRT}{MV}$. Using this in (ii), we get

$\Delta P = -\frac{mRT}{M} \quad (\because \Delta V = 2V - V = V)$

$\therefore \quad \Delta P_A = -\frac{m_A RT}{M}$ and $\Delta P_B = -\frac{m_B RT}{M}$

Hence $\frac{\Delta P_A}{\Delta P_B} = \frac{m_A}{m_B}$

@IITJEE_Advanced

Given $\Delta P_B = 1.5\Delta P_A$. Therefore, $\frac{1}{1.5} = \frac{m_A}{m_B}$
or $3m_A = 2m_B$, which is choice (c).

Q.9 Answer:

42 K

Solution:

Heat is given to the gas in cylinder A at constant pressure while the same amount of heat is given to the gas in cylinder B at constant volume. Heat given to gas in A is

$Q_A = nC_p\Delta T_A$

Heat given to gas in B is $Q_B = nC_v\Delta T_B$

Since $Q_A = Q_B$, we have

$nC_p\Delta T_A = nC_v\Delta T_B$

Or $\Delta T_B = \frac{C_p}{C_v}\Delta T_A$

$= \frac{7}{5} \times 30\ \text{K} = 42\ \text{K}$

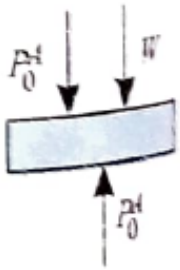
$(\because$ for a diatomic gas, $C_P/C_v = 7/5)$ Hence the correct choice is (d).

Q.10 Answer:

$\frac{2V_0}{7} + \frac{5nRT_0}{7\left(P_0 + \frac{W}{A}\right)}$

Solution:

The system is isolated from the surrounding, the work done by the gas will be at the cost of its own internal energy. Let the final temperature and volume be T and V, respectively. Let us suppose that the final pressure P of the gas will be just enough to hold the piston in equilibrium against its weight and the atmospheric pressure.



i.e., $PA = P_0^A + W$

$\Rightarrow \quad P = P_0 + \frac{W}{A}$

Change in interna energy of the gas,

$\Delta U = nC_V\Delta T = \frac{nR\Delta T}{(\gamma - 1)} = \frac{nR[T - T_0]}{\left(\frac{7}{5} - 1\right)} = \frac{5nR(T - T_0)}{2}$

$W = P\Delta'$

Work done by the gas against a constant pressure,

$\Rightarrow \quad W = \left(P_0 + \frac{W}{A}\right)(V - V_0)$

Using equation of state $PV = nRT$, we have

@IITJEE_Advanced

$\left(P_0+\frac{W}{A}\right)V=nRT$

$S=W+\Delta U=0\left[\because Q=0\right]$

$\Rightarrow\left(p_0+\frac{W}{A}\right)\left(V-V_0\right)+\frac{5nR(T-T_0)}{2}=0$ **...(iii)**

Solving eqn. (ii) and (iii), for V **and** T**,we get**

$T=\frac{5T_0}{7}+\left(P_0+\frac{W}{A}\right)\frac{2V_0}{7nR}$ **and** $V=\frac{2V_0}{7}+\frac{5nRT_0}{7\left(P_0+\frac{W}{A}\right)}$

Q.11 Answer:

3233.56 J

Solution:

Let P_1, V_1, T_1 **be the initiai pressure, volume and the temperature of the gas, respectively. For isothermal expansion (1 to 2), we have**

$V_2=2V_1; \quad P_1V_1=P_2V_2$

$P_2=P_1\left(V_1V_2\right)=P_1/2=0.5\times10^5\ \mathrm{N/m^2}$

$T_2=T_1=300\ \mathrm{K}$

For adiabatic expansion (2 to 3), $P_2V_2^\gamma=P_3V_3^\gamma$

$\Rightarrow \quad V_3=2V_2$

$\Rightarrow \quad P_3=P_2\left(\frac{V_2}{V_3}\right)^\gamma=0.5\times10^5\times(0.5)^{1.4}$

$\Rightarrow \quad P_3=1.9\times10^4\ \mathrm{N/m^2}$

$T_2V_2^{\gamma-1}=T_3V_3^{\gamma-1}$

$\Rightarrow \quad T_3=T_2\left(\frac{1}{V_3}\right)^{\gamma-1}=300\left(\frac{1}{2}\right)^{0.4}=227.35\ \mathrm{K}$

Hence, final pressure and temperatures are $1.9\times10^4\ \mathrm{N/m^2}$ **and** $227.35\ \mathrm{K}$**. respectively.**

Work done is $W=W_{1\to2}+W_{2\to3}$**.**

$W=2.303nRT_1\log\left(\frac{V_2}{V_1}\right)+\frac{nR(T_2-T_3)}{(\gamma-1)}$

$W=R(2.303\times300\times\log2)+\frac{R(300-227.35)}{0.4}$

$\Rightarrow W=3233.56\ \mathrm{J}$

Q.12 Answer:

$21\ \mathrm{J\ K^{-1}\ mol^{-1}}$

Solution:

Since the volume remains constant, $dW=0$**. Hence**

$dQ=dU=420\ \mathrm{J}$

$dQ=nC_vdT=2\times C_v\times10$

But $\left(\because dT=10^\circ\mathrm{C}=10\ \mathrm{K}\right)$

or $20C_v=420$ **or** $C_v=21\ \mathrm{J\ K^{-1}\ mol^{-1}}$**,**

which is choice (c).

Q.13 Answer:

$dU=0$

@IITJEE_Advanced

Solution:

In a cyclic process, the system returns to its initial state. Hence the change in internal energy $dU = 0$. Therefore, choice (b) is correct.

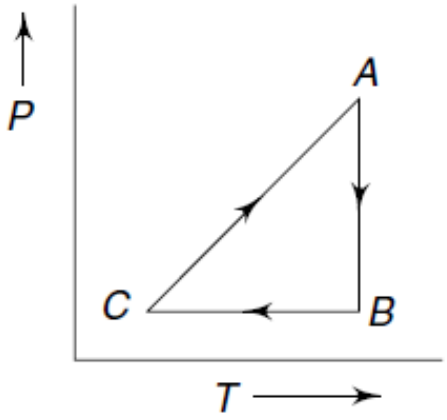
Q.14 Answer:

$\frac{T}{2}$

Solution:

Volume at A = Volume at $C = V$, but pressure at $A = 2P$ and at $C = P$. According to Charles' law, at constant volume, $P \propto T$. Hence the temperature at $A = \frac{T}{2}$, which is choice (d).

Q.15 Answer:



Solution:

Here the cycle is $ABCA$:

In P-V diagram:

- AB represents an isothermal process (hyperbolic).
- BC represents constant pressure process (isobaric).
- AC represents constant volume process (isochoric).

In P-T diagram

- AB will be a vertical straight line (isothermal process).
- BC will be a horizontal line (constant pressure process).
- AC will be a positive slope inclined line (constant volume process).

PV $= mRT$

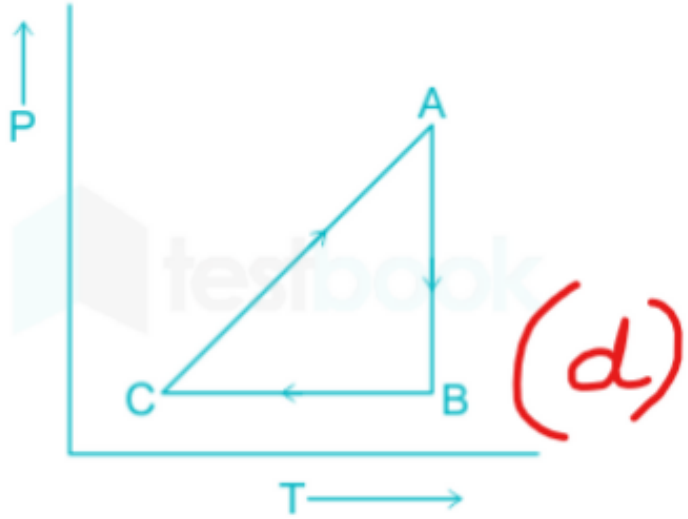
for ACV = constant

$\mathbf{P \propto T}$

$P = kT$ (Equation of line);

@IITJEE_Advanced

where k = constant



Q.16 Answer:

3452 J, 2764 J, 4980 J

Solution:

Let $P_1 = 1 \times 10^6$ Pa $= 10^6$ N/m^2; $T_1 = 300$ K; $n = 2$ moles.

Final volume $= 2$ (initial volume) $\Rightarrow V_2 = 2V$,

(a) $W_{\text{isothermal}} = 2.303nRT \log (V_2/V_1)$

$\Rightarrow \quad W = 2.303 \times 2 \times 8.3 \times 300 \log 2$

$W = 3452$ J

(b) $W_{(\text{adiabatic})} = \frac{nR(T_1 - T_2)}{\gamma - 1}$

For adiabatic process: $T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$

$T_2 = T_1 \left(\frac{V_1}{V_2}\right)^{\gamma-1} = 300 \times \left(\frac{1}{2}\right)^{(5/3)-1}$

$\Rightarrow \quad T_2 = 189$ K

$\dot{W} = \frac{nR(T_1 - T_2)}{\gamma - 1} = \frac{2 \times 8.3 \times (300 - 189)}{(5/3) - 1}$

$\Rightarrow \quad W = 2764$ J

(c) $W_{(\text{isobaric})} = P(V_2 - V_1) \Rightarrow P_1 V_1 = nRT_1$

$V_1 = \frac{nRT_1}{P_1} = \frac{2 \times 8.3 \times 300}{10^6}$

$\Rightarrow \quad V_1 = 4.98 \times 10^{-3}$ m^3

$W = P(V_2 - V_1) = P(2V_1 - V_1) = PV_1$

$\Rightarrow \quad W = 10^6 \times 4.98 \times 10^{-3}$ J $\Rightarrow W = 4980$ J

Q.17 Answer:

$n^{(1-\gamma)}$

Solution:

For isothermal process, PV = constant. Therefore

$P_i V_i = PV$

or $P_i \frac{V}{n} = PV$

Or $P_i = nP$

@IITJEE_Advanced

For adiabatic process, $PV^{\gamma} =$ constant. Therefore

$P_a\,(V_a)^{\gamma} = PV^{\gamma}$

or $P_a\left(\frac{V}{n}\right)^{\gamma} = PV^{\gamma}$

or $P_a = n^{\gamma}P$

From (i) and (ii) we get

$\frac{P_i}{P_a} = \frac{n}{n^{\gamma}} = n^{(1-\gamma)}$

Hence the correct choice is (d).

Q.18 Answer:

0.78 mm

Solution:

For an element dx at a distance x from the end P, the change in length

$dl = \alpha_1 dx(\theta - 10)$ or $\Delta l = \int dl$

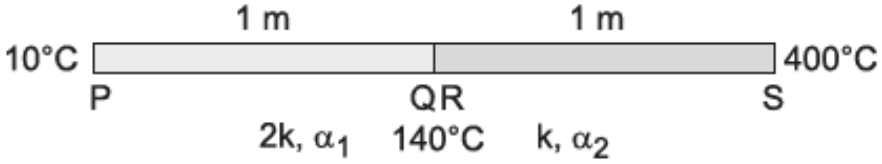
where θ is the temperature at the distance x from the end P.

Temperature gradient in $\mathrm{PQ} = \frac{\theta-10}{x} = \frac{140-10}{1}$

$\Rightarrow \theta = 130x + 10$

$$\Rightarrow \quad \Delta l = \int_0^1 (130x)\alpha_1 dx = 130\alpha_1 \frac{x^2}{2}\Big|_0^1 = 65 \times 1.2 \times 10^{-5}\ \mathrm{m}$$

$$= 78.0 \times 10^{-5}\ \mathrm{m} = 0.78\ \mathrm{mm}.$$



Q.19 Answer:

34%

Solution:

$T_1 = 227\ \ ^{\circ}\mathrm{C} = (227 + 273)\ \ \mathrm{K} = 500\ \ \mathrm{K}$

$T_2 = 57\ \ ^{\circ}\mathrm{C} = (57 + 273)\ \ \mathrm{K} = 330\ \ \mathrm{K}$

Efficiency of Carnot's engine,

$\eta = 1 - \frac{T_2}{T_1} = 1 - \frac{330}{500} = 0.34$ **or** $\eta = 34\%$

Q.20 Answer:

$400R\ln 2$

Solution:

@IITJEE_Advanced

In the process a to b and c to d

as $\Delta U = 0$, therefore $\Delta T = 0$ or $T =$ constant

$$W = \int_{V_i}^{V_f} P dV$$

We have,

$$PV = nRT \Rightarrow P = \frac{nRT}{V}$$

$$\therefore \quad W = \int_{V_i}^{V_f} nRT \frac{RV}{V}$$

$$= nRT ln \, \mathrm{V} \, {}_{V_i}^{V_f} = nRT \, \ln \frac{V_f}{V_i}$$

$$W_{ab} = nRT_b \, \ln \frac{2V_0}{V_0} = 2R \times 500 \, \ln 2 = 1000R \ln 2$$

and $W_{cd} = nRT_c \, \ln \frac{V_0}{2V_0} = 2R \times 300 \ln \frac{1}{2} = -600R \ln 2$

There is no volume change from b to c and from d to a, so

$$W_{bc} = W_{da} = 0$$

The work done in the complete cycle

$$W = W_{ab} + W_{bc} + W_{cd} + W_{da}$$

$$= 1000R \ln 2 + 0 - 600R \quad n2 + 0$$

$$= 400R \ln 2$$

For a cyclic process, the total work done and the total heat supplied are equal.

$\Rightarrow Q = 400R \quad \ln(2)$

@IITJEE_Advanced